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Modeling Helium Segregation to the Surfaces of Plasma-Exposed Tungsten as a Function of Temperature and Surface Orientation

Sophie Blondel,^{a,b} Karl D. Hammond,^{c,d} Lin Hu,^e Dimitrios Maroudas,^e and Brian D. Wirth^{a,f,*}

^aUniversity of Tennessee, Department of Nuclear Engineering, Knoxville, Tennessee 37996

^bOak Ridge National Laboratory, Computer Sciences and Mathematics Division, Oak Ridge, Tennessee 37831

^cUniversity of Missouri, Department of Chemical Engineering, Columbia, Missouri 65211

^dUniversity of Missouri, Nuclear Engineering Program, Columbia, Missouri 65211

^eUniversity of Massachusetts, Department of Chemical Engineering, Amherst, Massachusetts 01003-9303

^fOak Ridge National Laboratory, Fusion and Materials for Nuclear Systems Division, Oak Ridge, Tennessee 37831

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Abstract — We provide a description of the dependence on surface crystallographic orientation and temperature of the segregation of helium implanted with energies consistent with low-energy plasma exposure to tungsten surfaces. Here, we describe multiscale modeling results based on a hierarchical approach to scale bridging that incorporates atomistic studies based on a reliable interatomic potential to parameterize a spatially dependent drift-diffusion-reaction cluster-dynamics code. An extensive set of molecular dynamics (MD) simulations has been performed at 933 K and/or 1200 K to determine the probabilities of desorption and modified trap mutation that occurs as small, mobile He_n ($1 \leq n \leq 7$) clusters diffuse from the near-surface region toward surfaces of varying crystallographic orientation due to an elastic interaction force that provides the thermodynamic driving force for surface segregation. These near-surface cluster dynamics have significant effects on the surface morphology, the near-surface defect structures, and the amount of helium retained in the material upon plasma exposure, for which we have developed an extensive MD database of cumulative evolution during high-flux helium implantation at 933 K, which we compare to our properly parameterized cluster-dynamics model. This validated model is then used to evaluate the effects of temperature on helium retention and subsurface helium clustering.

Keywords — Helium clustering, multiscale modeling, plasma-surface interactions.

Note — Some figures may be in color only in the electronic version.

I. INTRODUCTION

Surface segregation of implanted species, such as helium (He) and hydrogen (H) isotopes, in plasma-facing components (PFCs) in nuclear fusion reactors underpins the key mechanisms of surface morphological evolution and near-surface structural evolution of these materials, as well as the retention and permeation of tritium.^{1,2} Tungsten (W) is a promising PFC material because its thermomechanical properties make it very appealing for functioning in such

extreme environments. Consequently, understanding the subsequent evolution of implanted helium atoms in the near-surface region of tungsten is required for the optimal design of PFCs to successfully sustain the extreme conditions under which they are expected to operate. This paper describes the continued establishment of benchmarks for hierarchical multiscale modeling tools to predict tungsten response to low-energy helium implantation, which is a necessary step toward using these tools to predict PFC performance in fusion reactors beyond ITER and to evaluate promising engineering designs for advanced PFCs.

*E-mail: bdwirth@utk.edu

It is well known that in linear plasma devices as well as tokamak plasmas, implanted helium atoms have significant impact on the PFC surface structure and morphology, causing the formation of microscopic features known as “fuzz” or “coral.”^{3–11} The dynamics of helium species in the PFC near-surface region mediate the formation of such complex surface morphologies; however, our understanding of the fuzz formation mechanism(s) is very limited. In tungsten, aggregation of mobile implanted interstitial helium atoms leads to formation of helium clusters, He_n , of varying sizes n (Refs. 12 through 18). The smallest of these helium clusters, with up to seven helium atoms ($n \leq 7$), are very mobile; the mobility of these clusters mediates their migration to the tungsten surface, facilitating helium surface segregation.^{19–21} As they approach the surface, these segregating small clusters may undergo a number of reactions that significantly impact the surface morphology and the near-surface structure and chemical composition.^{2,20,21} Computationally efficient quantitative prediction of the near-surface structural evolution and surface morphological evolution of PFCs requires a fundamental understanding and detailed modeling of helium transport and surface segregation in conjunction with the kinetics of helium cluster reactions that are enabled as helium segregates on the material surface.

The rest of this paper is structured as follows. The simulation methods employed in our study and their integration to model helium surface segregation and helium evolution in plasma-exposed tungsten are outlined in Sec. II. Our recent publication detailed the elastic interaction of small mobile helium (He_n , $1 \leq n \leq 7$) clusters with free surfaces in tungsten, which creates a driving force for the drift of helium toward the surface.^{2,19} In Sec. III, a detailed set of molecular dynamics (MD) simulations is analyzed to evaluate the probabilities of modified reactions and characterize their outcomes as small mobile helium (He_n , $1 \leq n \leq 7$) clusters migrate toward tungsten surfaces at either 933 or 1200 K, extending our prior studies.^{2,19,20} In Sec. IV, we present results from a continuum drift-diffusion-reaction rate theory based on a cluster-dynamics model, which is parameterized according to the findings of Sec. III, as well as systematic comparisons of the modeling results with those of large-scale MD simulations of cumulative helium implantation. Finally, the most important conclusions of this study are summarized in Sec. V.

II. SIMULATION METHODS

Our analysis of implanted helium transport toward surfaces of plasma-exposed tungsten follows a hierarchical, multiscale computational approach, which aims at (a) probing the interactions of helium-carrying species in

tungsten (small mobile helium clusters) with tungsten surfaces that underpin surface segregation and activate important kinetic phenomena in the near-surface regions, as well as characterizing these kinetic processes; (b) exploring directly the dynamics of helium segregation on surfaces and obtaining a fundamental understanding of this dynamics on the basis of the above-mentioned interactions and cluster kinetic processes; and (c) developing, parameterizing, and implementing computationally a continuum-scale model of helium cluster dynamics in the near-surface tungsten region capable of predicting helium bubble evolution in the material and determining the amount of helium retained in the tungsten. This approach bridges atomic-scale analysis of helium cluster–surface interactions and evolution of a helium population implanted in tungsten with continuum-scale modeling and simulation of cluster transport and reactions in the near-surface tungsten region.

Within this hierarchical computational framework, our simulation methodology integrates molecular statics (MS) computations of helium cluster energetics near tungsten surfaces, which have been presented in prior published papers^{2,19–21} and will not be discussed in great detail in this paper, with targeted MD simulations of cluster transport and reactions in the near-surface region, large-spatial-scale MD simulations of implanted helium aggregation and surface segregation, and continuum-scale self-consistent dynamical simulations of helium species (cluster) transport and reaction dynamics near free surfaces in tungsten. Our atomistic MD simulations were performed using the Large-scale Atomic/Molecular Massively Parallel Simulator²² (LAMMPS) open-source program. The MD simulations for analysis of near-surface cluster dynamics are detailed in Sec. III. The large-scale MD simulations of implanted helium evolution are described and analyzed in Sec. IV, and their results are compared to the predictions of our continuum-scale cluster-dynamics models, which account for both diffusive and advective (drift) transport for all mobile helium clusters coupled with nonlinear cluster reaction kinetics. The boundary value problems for the cluster concentration fields, consisting of the corresponding coupled nonlinear partial differential equations (PDEs) subject to proper boundary conditions, are solved numerically by implementing a centered finite-difference scheme that is described in an accompanying paper in this special issue of *Fusion Science and Technology*.²³ The parameterization of the continuum model is based entirely on our targeted atomistic simulations. The comparisons of the MD simulation results with the cluster-dynamics modeling predictions in Sec. IV focus on the helium distribution in the near-surface

region, helium retention in the material, and tungsten surface adatom generation under certain implantation conditions.

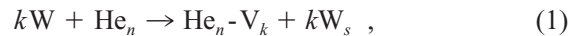
In our MS and MD simulations, the interactions between W atoms were described according to the potential developed by Finnis and Sinclair,²⁴ which is similar to the embedded atom method of Daw and Baskes²⁵ with embedding function $F(\rho) \propto \rho^{1/2}$, as modified at close range initially by Ackland and Thetford²⁶ and more recently by Juslin and Wirth.²⁷ We have adopted the description of W-He pairwise interactions according to the pair potential of Juslin and Wirth²⁷ and of He-He interactions according to the pair potential developed by Beck^{28,29} as modified at short distances by Morishita et al.³⁰ Simulations based on these interatomic interaction potentials have reproduced accurate results for the formation energies and ground-state configurations of He point defects and self-interstitial atoms in W obtained by first-principles density functional theory (DFT) calculations.^{27,31} The binding energies of small helium-vacancy clusters predicted by these potentials are in excellent qualitative agreement with the DFT results; however, the potentials overestimate slightly but systematically vacancy binding energies.³¹ In previous computational studies, helium implantation effects on tungsten surface evolution were explored in MD simulations based on these interatomic potentials.^{2,12,14–21,32,33} In general, the interatomic potentials employed in this study are considered to be very reliable for analysis of the energetics and dynamics of small mobile helium clusters in tungsten.

III. ANALYSIS OF HELIUM CLUSTER–SURFACE INTERACTIONS AND HELIUM CLUSTER NEAR-SURFACE REACTIONS

A helium cluster in the tungsten matrix constitutes a center of dilatation that induces atomic displacements in the surrounding tungsten atoms, is highly mobile, and strongly interacts via its atomic displacement field with free surfaces as described in our recent publications based on elastic inclusion theory in conjunction with systematic MS simulations.^{2,19} We also analyzed the near-surface reactions undergone by these small mobile He_n clusters ($1 \leq n \leq 7$) in tungsten as a function of surface orientation through a systematic MD simulation study of cluster dynamics for all the mobile He_n clusters in tungsten near W(100), W(110), W(111), and W(211) surfaces at temperatures of 933 and 1200 K (Refs. 19 and 21). For each cluster examined ($1 \leq n \leq 7$) and each surface orientation, several hundred trajectories were generated, each of which was several nanoseconds long (5 to 10 ns, depending on cluster size and surface orientation), at 933 K or 1200 K. Further details regarding the simulations are provided in

Refs. 19 and 21, with the outcomes of this analysis used as input to the continuum-scale cluster-dynamics simulations discussed in Sec. IV.

The cluster kinetic processes identified by the MD analysis are trap mutation (TM), cluster partial dissociation (PD) following TM, and helium desorption from the surface.^{19–21} For $n > 1$, helium escapes back into the plasma because of cluster dissociation. Trap mutation, which is the process of a growing helium cluster or bubble causing the ejection of one or more metal atoms as interstitial(s), leads to the formation of one or more ($k \geq 1$) tungsten vacancies in lattice planes near the surface plane together with an equal number (k) of tungsten adatoms (atoms “on the surface”). The He_n cluster becomes attached to the generated vacancies, forming an immobile subsurface He_n -vacancy complex. In the bulk material, TM reactions produce Frenkel pairs corresponding to equal numbers of vacancies and self-interstitial W atoms instead of tungsten adatoms; such TM reactions, involving Frenkel pair formation in the bulk metal, are typically characterized by high activation energies. The near-surface TM reaction can be expressed as



where V and W_s denote a lattice vacancy and a tungsten adatom, respectively. The sequential reaction of TM followed by PD is expressed as



with $m < n$ and $\text{He}_{(g)}$ denoting a helium atom that escapes from the tungsten surface to the gas phase. For $m = 1$ and $k = 1$, $\text{He}_m\text{-V}_k$ corresponds to a substitutional helium atom; for $m > 1$, it corresponds to a bubble (helium trapped by a vacancy). For $m = 0$, reaction (2) corresponds to $W + \text{He}_n \rightarrow W + n\text{He}_{(g)}$, i.e., complete dissociation of the cluster leading to the release of all the helium in the cluster from the tungsten surface.

Trap mutation is by far the most important of these reactions; i.e., the corresponding reaction probability renders it the dominant cluster reaction for most cluster sizes and surface orientations studied. Specifically, we found TM reactions to be dominant for all combinations of cluster sizes and surface orientations, with the exception of the He_4 and He_5 clusters near the W(100) surface²¹; for these combinations, cluster PD reactions are the dominant ones, although it should be emphasized that PD reactions always occur as steps in a sequential reaction process that occurs subsequent to a TM reaction.^{19–21} The reason why TM reactions are the dominant kinetic processes in He_n cluster dynamics near W(110), W(111), and W(211)

surfaces are the low activation energies for TM reactions in conjunction with the high energy barriers for PD reactions near these surfaces. We also found that there is a critical cluster size, $n_c = 4$ to 5, for the formation of multiple ($k > 1$) tungsten adatoms and vacancies in TM reactions,²¹ i.e., $k = 1$ for $n < n_c$; $n_c = 4$ near W(100), W(111), and W(211) and $n_c = 5$ near W(110) with values of k up to 4 for a single TM reaction near W(100), W(111), and W(211) surfaces.

Tables I through IV provide a detailed assessment of the observed reaction outcomes for the mobile helium clusters (He_n , $1 \leq n \leq 7$) as they diffuse toward the W(100), W(110), W(111), and W(211) surfaces, respectively, at 933 and 1200 K. They correspond to a summarized version of the complete list of reactions, in which only the dominant reaction for each cluster size is provided. In Tables I through IV, the first column is the number of helium atoms contained in the cluster, the second column is the atomic layer at which the reaction occurs, the third column is the corresponding depth from the surface (provided since the distance between atomic layers varies with the surface orientation), the fourth column lists the outcome of the TM reaction (e.g., how many interstitials are generated), and finally the last column provides the percentage of the helium atoms from the

cluster that undergo desorption in the MD simulation. Notably, the probability of helium escaping slightly increased with increasing temperature from 933 to 1200 K for the W(100) and W(110) surfaces, whereas the desorption probability decreased with increasing temperature for the W(111) and W(211) surfaces. Presumably, this is due to the reaction dynamics associated with overcoming the activation barrier(s) for TM at the higher temperature near the W(100) and W(110) surfaces, which are higher than those near the W(111) and W(211) surfaces. The outcomes listed in Tables I through IV have been fully implemented in our one-dimensional drift-diffusion-reaction cluster-dynamics model, as has the elastic interaction force responsible for the cluster drift fluxes toward the material surfaces.

IV. CLUSTER-DYNAMICS MODELING OF HELIUM TRANSPORT, AGGREGATION, AND REACTIONS AS A FUNCTION OF SURFACE ORIENTATION

The continuum-scale cluster-dynamics simulation program Xolotl has been developed to efficiently integrate numerous coupled PDEs, namely, the one-dimensional drift-diffusion-reaction equations for the evolution of the concentration fields of interstitial helium clusters, as well

TABLE I

Observed Reaction Outcomes and Computed Reaction Probabilities from MD Simulations of Mobile Helium Cluster Dynamics Near the W(100) Surface at Temperatures of 1200 and 933 K

$T = 1200 \text{ K}$				
Cluster Size	Layer	Depth (nm)	Products	Desorption
2	3	0.5	$\text{He}_2\text{V} + \text{I}$	19%
3	4	0.6	$\text{He}_3\text{V} + \text{I}$	—
4	5	0.8	$\text{He}_4\text{V} + \text{I}$	—
5	4	0.6	$\text{He}_5\text{V}_2 + \text{I}_2$	—
6	5	0.8	$\text{He}_6\text{V}_2 + \text{I}_2$	—
7	5	0.8	$\text{He}_7\text{V}_2 + \text{I}_2$	—
$T = 933 \text{ K}$				
Cluster Size	Layer	Depth (nm)	Products	Desorption
2	3	0.5	$\text{He}_2\text{V} + \text{I}$	4%
3	4	0.6	$\text{He}_3\text{V} + \text{I}$	—
4	4	0.6	$\text{He}_4\text{V} + \text{I}$	—
5	5	0.8	$\text{He}_5\text{V} + \text{I}$	—
6	5	0.8	$\text{He}_6\text{V}_2 + \text{I}_2$	—
7	5	0.8	$\text{He}_7\text{V}_2 + \text{I}_2$	—

TABLE II

Observed Reaction Outcomes and Computed Reaction Probabilities from MD Simulations of Mobile Helium Cluster Dynamics Near the W(110) Surface at Temperatures of 1200 and 933 K

$T = 1200 \text{ K}$				
Cluster Size	Layer	Depth (nm)	Products	Desorption
2	3	0.7	$\text{He}_2\text{V} + \text{I}$	32%
3	4	0.9	$\text{He}_3\text{V} + \text{I}$	—
4	4	0.9	$\text{He}_4\text{V} + \text{I}$	—
5	4	0.9	$\text{He}_5\text{V} + \text{I}$	—
6	5	1.1	$\text{He}_6\text{V} + \text{I}$	—
7	5	1.1	$\text{He}_7\text{V}_2 + \text{I}_2$	—
$T = 933 \text{ K}$				
Cluster Size	Layer	Depth (nm)	Products	Desorption
2	3	0.7	$\text{He}_2\text{V} + \text{I}$	31%
3	4	0.9	$\text{He}_3\text{V} + \text{I}$	—
4	4	0.9	$\text{He}_4\text{V} + \text{I}$	—
5	4	0.9	$\text{He}_5\text{V} + \text{I}$	—
6	5	1.1	$\text{He}_6\text{V}_2 + \text{I}_2$	—
7	5	1.1	$\text{He}_7\text{V}_2 + \text{I}_2$	—

TABLE III

Observed Reaction Outcomes and Computed Reaction Probabilities from MD Simulations of Mobile Helium Cluster Dynamics Near the W(111) Surface at Temperatures of 1200 and 933 K

T = 1200 K				
Cluster Size	Layer	Depth (nm)	Products	Desorption
1	6	0.6	HeV + I	35%
2	9	0.8	He ₂ V + I	—
3	12	1.1	He ₃ V + I	—
4	12	1.1	He ₄ V + I	—
5	12	1.1	He ₅ V + I	—
6	12	1.1	He ₆ V + I	—
7	12	1.1	He ₇ V ₂ + I ₂	—
T = 933 K				
Cluster Size	Layer	Depth (nm)	Products	Desorption
1	6	0.6	HeV + I	61%
2	9	0.8	He ₂ V + I	—
3	12	1.1	He ₃ V + I	—
4	12	1.1	He ₄ V + I	—
5	13	1.2	He ₅ V + I	—
6	15	1.3	He ₆ V + I	—
7	15	1.3	He ₇ V ₂ + I ₂	—

as a large network of helium-vacancy clusters and small tungsten self-interstitial atom clusters. The software performs continuum-domain discretization using a finite-difference grid and approximates the spatial derivatives using centered finite-difference expressions, as described in a recent publication² and the accompanying paper²³ that describes a multidimensional version of the code. An initial benchmark of Xolotl against a set of results from large-scale MD simulations of the accumulation kinetics of helium clusters during high-flux implantation of 100-eV helium atoms into single crystals of tungsten as a function of surface orientation at 933 K has been recently published,² based on a systematic parameterization of helium cluster elastic interactions with tungsten surfaces and modified reactions near surfaces. The helium cluster elastic interactions were implemented in the cluster-dynamics code through a biased drift term as previously detailed in Ref. 2, while the modified reactions near the surface follow the list in Tables I through IV such that the rate, or kinetic prefactor, are set 1000 times higher than the highest rate of any other reaction. In this section, we provide a more detailed benchmark assessment prior to evaluating the effect of temperature on the segregation of

TABLE IV

Observed Reaction Outcomes and Computed Reaction Probabilities from MD Simulations of Mobile Helium Cluster Dynamics Near the W(211) Surface at Temperatures of 1200 and 933 K

T = 1200 K				
Cluster Size	Layer	Depth (nm)	Products	Desorption
1	4	0.5	HeV + I	59%
2	6	0.8	He ₂ V + I	—
3	8	1.0	He ₃ V + I	—
4	8	1.0	He ₄ V ₂ + I ₂	—
5	10	1.3	He ₅ V + I	—
6	10	1.3	He ₆ V ₂ + I ₂	—
7	9	1.2	He ₇ V ₃ + I ₃	—
T = 933 K				
Cluster Size	Layer	Depth (nm)	Products	Desorption
1	4	0.5	HeV + I	64%
2	6	0.8	He ₂ V + I	—
3	8	1.0	He ₃ V + I	—
4	8	1.0	He ₄ V ₂ + I ₂	—
5	8	1.0	He ₅ V ₂ + I ₂	—
6	10	1.3	He ₆ V ₂ + I ₂	—
7	12	1.5	He ₇ V + I	—

helium on tungsten surfaces and the subsurface helium cluster populations.

In these MD simulations, helium is implanted directly below the surface according to a predetermined depth distribution, which is different for each surface crystallographic orientation examined and depends strongly on the impinging ion energy, as described in our previous publication.³² In the subsequent large-scale MD simulations of helium evolution (aggregation and surface segregation) in the near-surface region of the material, the top boundary of the simulation supercell was treated as a free surface, the bottom surface was frozen at its crystallographic position to prevent permeation, and periodic boundary conditions were applied in the principal directions of the horizontal surface plane; the dimensions of the simulation supercell employed in this study were ~50 × 50 × 25 nm, with the exception of the W(111) simulation, in which the dimensions of the simulation cell were ~50 × 50 × 21 nm. Consistent with the studies of Sec. III, the surfaces examined were W(100), W(110), W(111), and W(211). The implanted helium atoms were inserted every 10 ps at random horizontal locations and at depths sampled from the depth distribution characteristic of the

surface's orientation; this rate of insertion and simulation parameters correspond to a nominal helium flux of $4.0 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$, neglecting reflected ions (i.e., this flux is the flux of helium ions that embed below the surface). Temperature control in these simulations was performed with a Nosé-Hoover chain thermostat.^{34–36} The implementation details of the helium insertion process and temperature control were consistent with those of our previous publication.^{18,37}

Figure 1 shows a comparison of the Xolotl predictions for helium retention below a W(211) surface at 933 K, which provides an update to the comparison presented in our earlier publication.² In that work, the atomistic database on the modified reactions near the W(211) surface had relied on data obtained for the W(111) surface. Here, we have completed the systematic MD evaluation of the modified helium cluster reactions at both 1200 K, which was the temperature at which the database used in Ref. 2 was obtained, and 933 K. Figure 1 shows the impact of properly changing the database for the modified helium cluster reactions near the W(211) surface on the predictions obtained by Xolotl for the helium retention in the material as a function of fluence at 933 K for 100-eV helium implantation at a nominal flux of $4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$. As shown by the solid line, the influence of

incorporating the database of modified reactions specific to the W(211) surface is to lower the overall level of retained helium compared to that obtained according to the database specific to the W(111) surface. This can be understood by considering the probabilities listed in Tables III and IV, in which the probability for desorption of a single helium atom at 1200 K is increased from 35% to 59% for the W(211) surface relative to that for the W(111) surface. Furthermore, the depth at which the modified TM reactions are observed to occur is slightly deeper (in layer number) for the reactions occurring below the W(111) surface than those occurring below the W(211) surface.

Our earlier work comparing large-scale MD simulation results to predictions from Xolotl calculations was performed at 933 K based on a comprehensive database of near-surface cluster reaction probabilities obtained from targeted MD simulations detailing the modified reactions at 1200 K; this was necessary because we lacked (at the time) a database of probabilities at 933 K. To understand the impact of this temperature shift in the reaction database used as input to the Xolotl calculations, we have now extended the atomistic database to 933 K, as demonstrated in Tables I through IV. Figure 2 shows the impact of this temperature difference between 933 and 1200 K on the predictions of helium retention in the material as a function of helium fluence from the continuum-scale Xolotl calculations. In particular, Fig. 2a shows the impact of using the atomistic database obtained at 1200 K (dashed lines) versus that obtained at 933 K (solid lines) on helium retention as a function of surface orientation at 933 K. Notably, the use of the atomistic database obtained at the correct temperature (consistent with that of the large-scale MD simulations) decreased the overall helium retention predicted by Xolotl for the W(111) and W(211) surfaces, while the predicted retention increased for the W(100) surface and remained largely unchanged for the W(110) surface. As discussed previously with respect to the results in Tables I through IV, the accumulated retention kinetics as a function of surface orientation matches very well with the results of the targeted MD simulation database of near-surface cluster reactions in which the probability of helium desorption from the surface slightly increased in going from 933 to 1200 K for the W(100) surface and was essentially unchanged for the W(110) surface, whereas the desorption probability decreased with increasing temperature for the W(111) and W(211) surfaces. Correspondingly, the differences in the modified reaction database from 1200 to 933 K are directly mapped onto the accumulation and retention behavior of helium in the material during continuous plasma exposure.

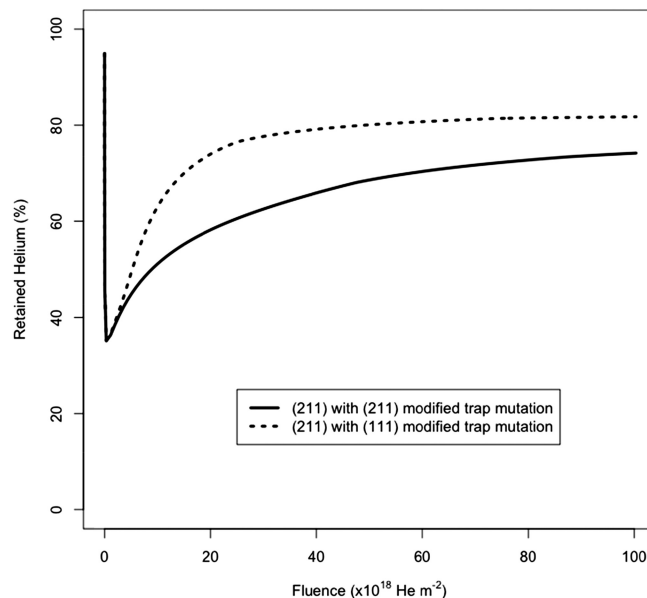


Fig. 1. Helium retention below a W(211) surface as a function of fluence predicted by Xolotl at 933 K for 100-eV helium implantation at a nominal flux of $4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$. The dashed line is an approximate result, for which we used the modified rates of mobile helium cluster reactions near the W(111) surface, whereas the solid line now incorporates the parameterization from MD simulations of mobile helium cluster reactions near the W(211) surface described in Table IV.

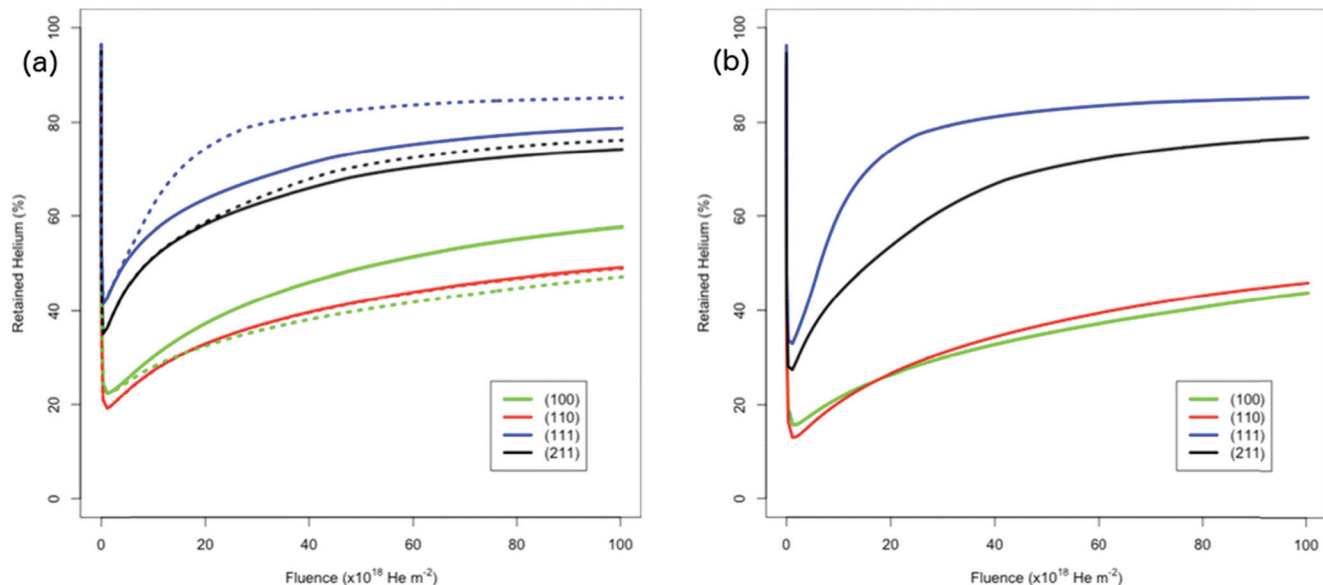


Fig. 2. Helium retention below the W(100), W(110), W(111), and W(211) surfaces as a function of implanted helium fluence predicted by Xolotl at (a) 933 K and (b) 1200 K for 100-eV helium implantation at a nominal flux of $4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$. In (a), the dashed lines show the impact of using the parameterization for modified helium cluster reactions near the indicated surfaces obtained at 1200 K in the 933 K simulations, whereas the solid lines are based on the atomistic database obtained at 933 K.

A recent experimental study³⁸ reported helium concentrations under various tungsten surfaces, including “fuzzy” surfaces, as a function of surface temperature and helium fluence, with significantly lower helium implantation flux than used in our current simulation study. Over the temperature range of 470 to 2595 K and the helium fluence range of 3×10^{24} to $1.3 \times 10^{27} \text{ m}^{-2}$ examined, the measured helium concentration was found to be in the range of 1 to 4 at. % and largely independent of temperature.³⁸ Thus, our modeling results presented in Fig. 2 support the lack of a strong temperature dependence, but there is clearly a strong implantation rate effect on the absolute magnitude of the fraction of retained helium, which will require further modeling work to resolve. We do believe that the lack of a strong temperature dependence in helium retention is consistent with our computational findings: The dominant, low-barrier TM reactions retain helium in the near-surface region of tungsten, trapped in the form of He_mV_k complexes, as the small mobile helium clusters undergo drift transport toward the surface.

Figure 3 shows the impact of this temperature difference between 933 and 1200 K on the continuum Xolotl calculations of the cumulative helium distribution. In particular, Figs. 3a and 3c show the impact on the helium depth distribution as a function of surface orientation at 933 K by using the atomistic database specific to 1200 K (dashed lines) versus using the more accurate one specific to the actual temperature of 933 K (solid lines). As noted in our previous publication,² the results shown in Fig. 3

clearly identify a near-surface segregation region of helium, which is in reasonable quantitative agreement with the large-scale MD simulations at the lower-fluence values.² The differences in the He distribution at different surface orientations are explained well from the MD analyses presented previously,^{19–21,32} since the W(111) and W(211) surfaces are stronger sinks for mobile He_n clusters than the W(100) and W(110) surfaces, resulting in more effective helium surface segregation on the former two surfaces. The differences in the strength of the elastic driving force for surface segregation is supplemented by the modified TM reactions near surfaces that are presented in Tables I through IV. At low fluence, the effect of temperature on the cumulative distribution of helium in the material below the surface is relatively minor, but some differences appear at higher fluences. However, as noted in our previous study,² the cumulative depth distributions predicted by the Xolotl calculations do not agree as well with the large-scale MD simulations, in which the near-surface segregation plateau of the helium distribution is no longer clearly observable. This discrepancy is attributed to several factors, including the possibility that the modified TM reactions are dependent on the local concentrations of He_n clusters, which serve as sinks for mobile helium and drive helium bubble growth.

Figure 4 shows a comparison between the Xolotl predictions (dashed lines) of total helium retention at 933 K with the results of large-scale MD simulations (solid lines) of helium clustering and aggregation kinetics. Compared with the early-stage helium depth distribution

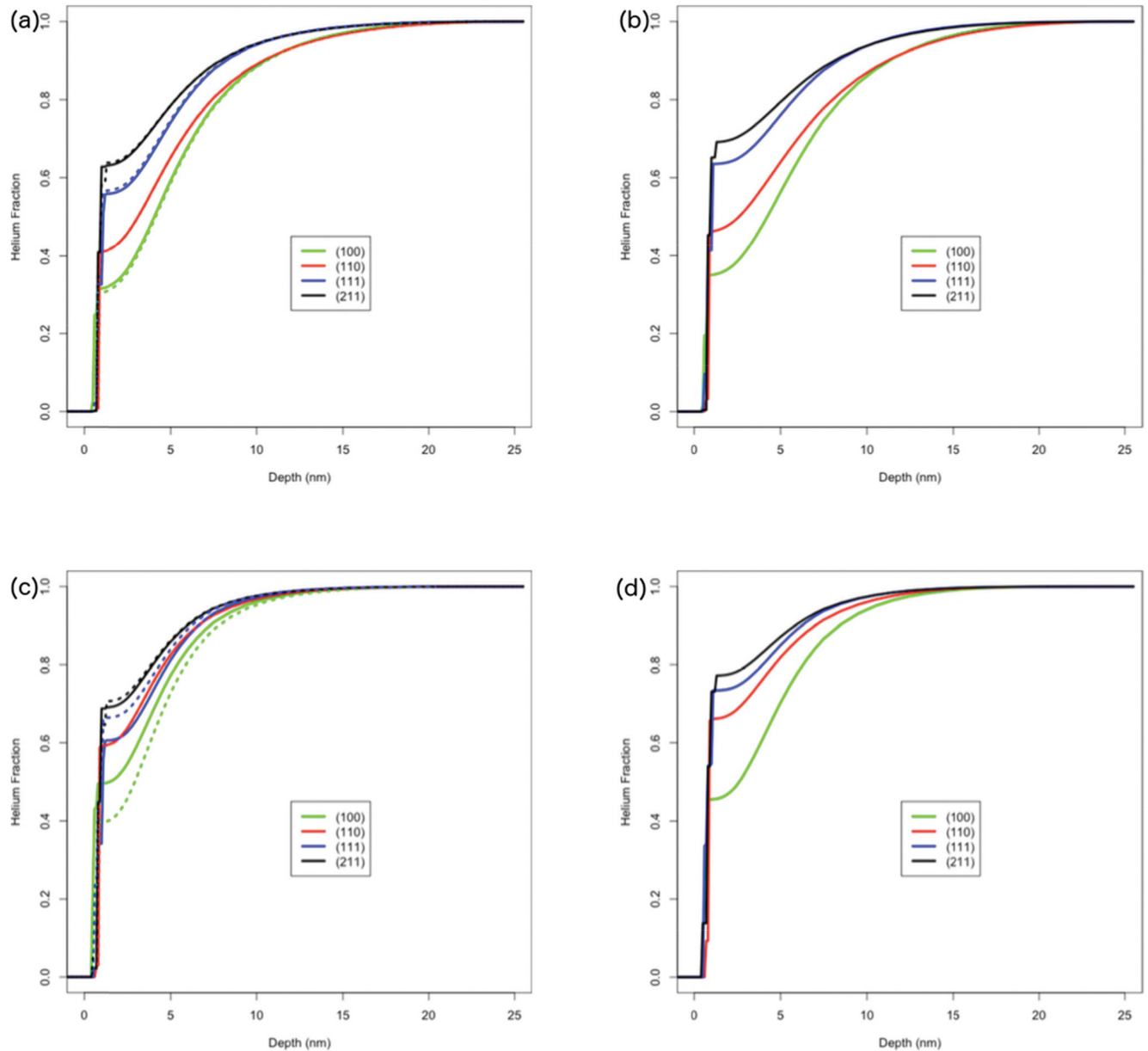


Fig. 3. Cumulative distribution of helium as a function of depth below the surface for several surface orientations during 100-eV helium implantation for a nominal flux of $4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$ at (a) 933 K at a nominal fluence of $\sim 3 \times 10^{18} \text{ m}^{-2}$, (b) 1200 K at a nominal fluence of $\sim 3 \times 10^{18} \text{ m}^{-2}$, (c) 933 K at a nominal fluence of $\sim 3 \times 10^{19} \text{ m}^{-2}$, and (d) 1200 K at a nominal fluence of $\sim 3 \times 10^{19} \text{ m}^{-2}$. In (a) and (c), the dashed lines are the results corresponding to an atomistic database of modified helium cluster reactions near the indicated surfaces obtained at 1200 K, whereas the solid lines are the results from the atomistic MD database obtained at 933 K.

presented in our previous paper,² the current comparison is not as good quantitatively, and except for the lower fluences, the Xolotl continuum model predicts higher levels of retained helium than those observed in the MD simulations. In addition, the MD simulations indicate a higher level of helium retention for the W(211) surface than the W(111) surface, which is the opposite of the Xolotl predictions. There is better quantitative agreement between the Xolotl predictions and the MD simulation

results for the W(100) and W(110) surfaces in terms of the absolute retention values and the relative differences in helium retention between the two surface orientations, although the continuum model predicts a slightly lower retention at low fluence but overpredicts the retention at higher fluences. It should be mentioned that one important aspect of the MD simulations that has not yet been implemented into the continuum model is the bursting of helium bubbles that are close to the tungsten surface,

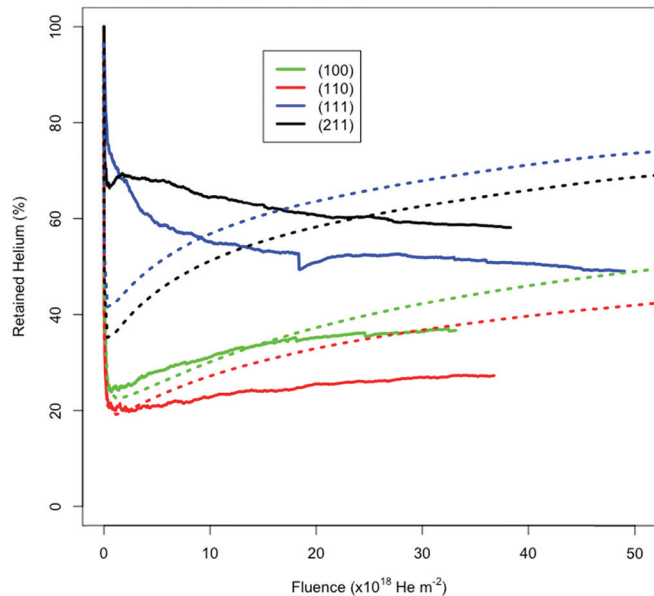


Fig. 4. Comparison of MD simulation results (solid lines) with Xolotl predictions (dashed lines) for helium retention as a function of helium fluence at 933 K during 100-eV helium implantation at a flux of $\sim 4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$.

which results in a significant release of helium, from the helium contained in the bubble, with a corresponding sharp decrease in the helium retention in the material that is observed at a nominal fluence of $\sim 18 \times 10^{18} \text{ He m}^{-2}$ for the W(111) surface. We are currently working on implementing a simplified model for this bubble bursting process, based on the distance between the bubble and the surface, as well as considering a model informed by MD simulations that evaluate the critical bubble pressure as a function of distance below the surface, but we are not yet able to report any significant progress.

Additional benchmarks of the Xolotl modeling capabilities can be obtained through comparing its predictions for the helium cluster size distributions as a function of depth below the tungsten surface with the corresponding results from the large-scale MD simulations. Figures 5, 6, and 7 provide comparisons between the MD and Xolotl predictions of the helium concentration as a function of depth and fluence.

More specifically, Fig. 5 shows the total volumetric concentration of helium, defined by the number of helium atoms in a cluster of given size times the density of the cluster, as a function of depth at different fluences for MD and Xolotl simulations. The peak near the surface is due to the TM processes described in Sec. III. Xolotl is able to reproduce this surface segregation effect, but clearly overestimates the quantity of helium in the material at higher fluence. One approach to overcoming this problem is to attenuate the effect of TM near the surface as a function of

helium segregation, cluster density, or time. Indeed, all the parameters described in Sec. III were obtained from MD simulations that did not contain pre-existing helium clusters in the tungsten material. One can assume that once the helium clusters begin to saturate the free surface, the sink strength will be gradually reduced. Thus we have incorporated such an effect in Xolotl by modifying the TM rate near the surface as

$$k'_{TM} = k_{TM} e^{-A C_{\text{He}}} , \quad (3)$$

where k_{TM} and k'_{TM} are the original and modified TM rates, respectively, C_{He} is the local total helium concentration, and A is an adjustable parameter. The dotted line in Fig. 5 shows Xolotl results with the inclusion of this attenuation factor in the corresponding reaction rates. The agreement between MD and Xolotl results is improved, but Xolotl still overestimates the overall helium retention at high fluence.

It is important to note that this attenuation factor is an early, and essentially *ad hoc* addition to Xolotl that is not yet based on data from targeted MD simulations. We believe such an effective screening of the elastic interactions resulting from the presence of multiple bubbles and small helium-vacancy clusters can explain this discrepancy, because the MD simulations used to populate the modified TM reactions provided in Tables I through IV were all performed in the dilute limit. We have now initiated additional MD simulations, and we expect to be able to report in the future regarding a more systematic evaluation of this proposed difference between the large-scale MD and continuum Xolotl results.

The helium cluster size distribution is shown in Fig. 6 as a function of depth and helium cluster size, as computed by the large-scale MD simulations and predicted by Xolotl at a fluence of $1 \times 10^{19} \text{ m}^{-2}$. Both simulations are in good qualitative agreement, and we believe that the lack of statistics from the MD simulations is likely the source of the scattered data in Fig. 6a. Finally, Fig. 7 is similar to Fig. 5, except that the concentration of helium and helium clusters is plotted as a function of depth and implanted helium fluence, whereas Fig. 5 shows the total concentration of helium atoms. From Fig. 7, it seems clear that Xolotl is predicting an appropriate spatial distribution of bubbles (with the use of the attenuation factor described above) even at high fluence, although the size distribution of the helium bubbles is not yet correct, as can be seen in Fig. 6, where Xolotl predicts larger bubble sizes near the surface where the concentration of helium is the highest. We believe this to be the cause of the discrepancies in Fig. 5. We think this effect might be related to conditions where the dilute limit is no longer applicable and where the reaction rates would be overpredicted, but

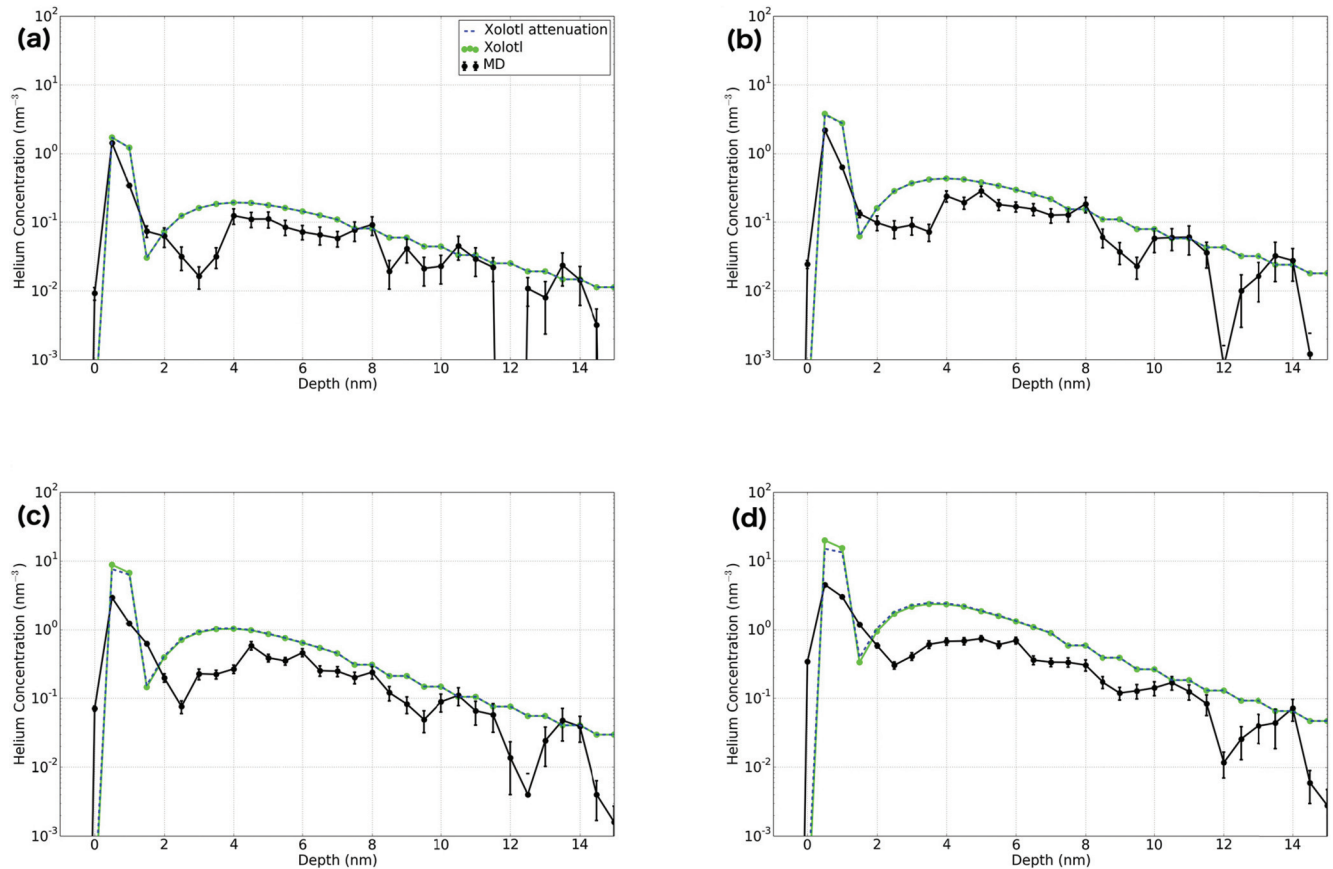


Fig. 5. Comparison of MD simulation results (solid circles with accompanying error bars from statistical analysis of MD trajectories) with Xolotl predictions without attenuation (solid circles) and with attenuation (dotted lines) of near-surface reaction rates for helium concentration as a function of depth below a W(111) surface at 933 K during 100-eV helium implantation at a flux of $\sim 4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$, with a helium implantation fluence of (a) $5 \times 10^{18} \text{ m}^{-2}$, (b) $1.0 \times 10^{19} \text{ m}^{-2}$, (c) $2.0 \times 10^{19} \text{ m}^{-2}$, and (d) $4.0 \times 10^{19} \text{ m}^{-2}$.

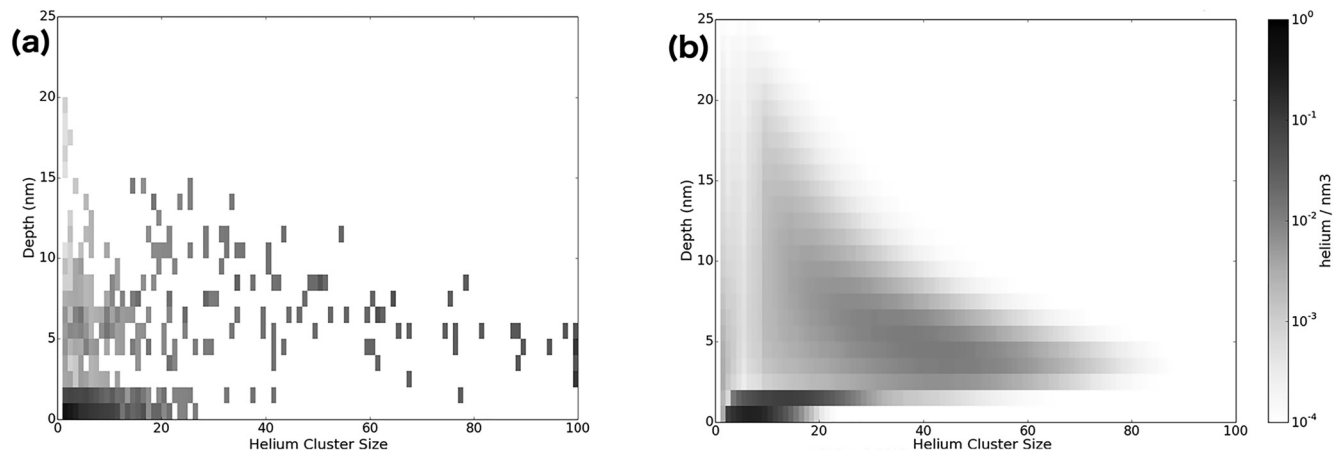


Fig. 6. Comparison of (a) MD simulation results with (b) Xolotl predictions for the volumetric helium density as a function of helium cluster size and depth below a W(111) surface at 933 K during 100-eV helium implantation at a flux of $4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$ to a fluence of 10^{19} m^{-2} .

more work is required to investigate these discrepancies, and this work is beyond the scope of the present paper as stated in our accompanying paper.²³

A comparison of the adatom areal density predicted by MD simulations and by Xolotl as a function of time is shown in Fig. 8 for different surface orientations at 933 K

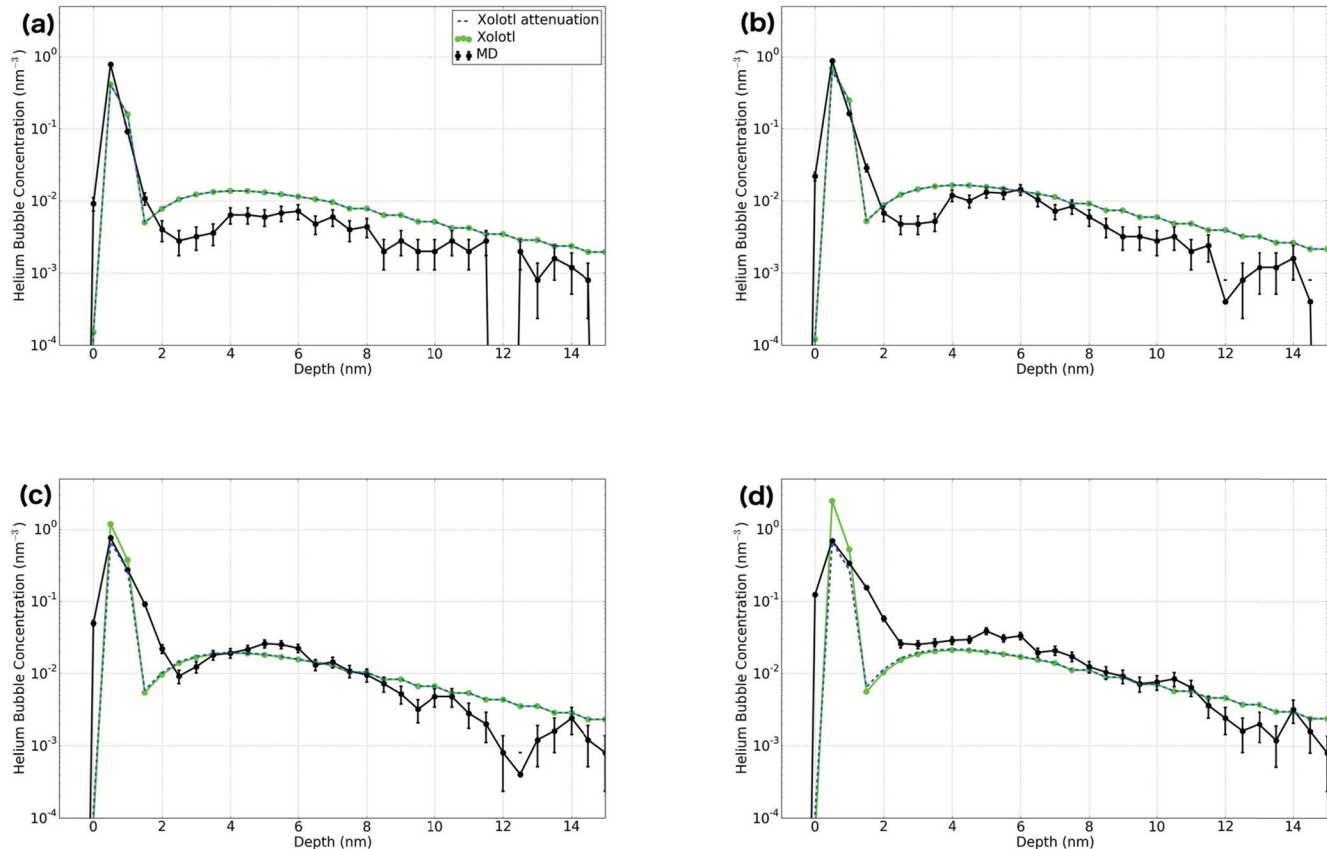


Fig. 7. Comparison of MD simulation results (solid circles with accompanying error bars from statistical analysis of the MD trajectories) with Xolotl predictions without attenuation (solid circles) and with attenuation (dotted lines) of near-surface reaction rates for helium bubble concentration as a function of depth below a W(111) surface at 933 K during 100-eV helium implantation at a flux of $\sim 4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$, with a helium implantation fluence of (a) $5 \times 10^{18} \text{ m}^{-2}$, (b) $1.0 \times 10^{19} \text{ m}^{-2}$, (c) $2.0 \times 10^{19} \text{ m}^{-2}$, and (d) $4.0 \times 10^{19} \text{ m}^{-2}$.

during 100-eV helium implantation at a nominal flux of $4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$. The main processes generating self-interstitials are TM and loop punching, which occur when the helium gets trapped and begins to grow into overpressurized bubbles below the tungsten surface. The agreement at short times for the (100) and (110) surfaces between MD and continuum predictions is good, but the predicted areal density diverges at longer times, presumably due to the increased contribution of prismatic loop punching to the adatom density, which is not yet incorporated into Xolotl. The results from MD simulation for the (111) and (211) surfaces are both surprising, as one would expect the adatom areal density to be roughly proportional to the helium retention, since adatoms are easily created by mobile self-interstitial tungsten atoms segregating to the free surface and undergoing the modified TM for these surfaces.³² Xolotl also predicts this behavior, as the differences among the surface orientations are sorted in the same order as the retention results presented in Fig. 4. This difference probably means that there are some processes that participate in the adatom

formation and are active in the MD simulations that have not yet been incorporated into our coarse-grained cluster-dynamics model.

It should be noted that exactly what constitutes an adatom is not nearly as straightforward in MD simulations as it is in continuum models such as Xolotl that label atoms as being in distinct layers. In an MD simulation, as in real materials, the atoms near the surface are constantly vibrating, meaning a simplified analysis using solely the position of the atoms above the original surface (with no added margin) will yield a large number of false positives in identifying adatoms. This is further complicated by the fact that the difference between subsequent layers of atoms is quite different between the (100), (110), (111), and (211) surfaces (see one of our previous publications³² for a quantitative comparison). A threshold of just under 0.5 lattice parameters might make sense for a (100) surface, for example, but the (111) surface has two atoms in staggered positions that would be counted as false negatives using the same threshold. The plots discussed here use a threshold of 0.159 nm, corresponding to $0.5a$, where

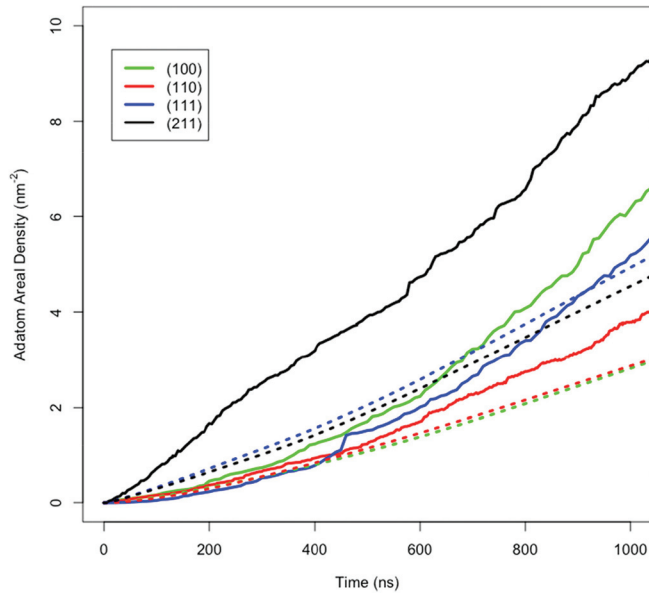


Fig. 8. Comparison of tungsten adatom areal density, resulting from the transport of tungsten self-interstitial atoms and prismatic dislocation loops to the surface, as a function of time, comparing MD simulation results (solid lines) with Xolotl predictions with the attenuation of near-surface reaction rates (dashed lines) for four different surface orientations at a flux of $4 \times 10^{25} \text{ m}^{-2} \cdot \text{s}^{-1}$.

a is the lattice parameter, for the identification of adatoms from MD simulations. Decreasing the threshold to 0.1 nm (approximately $a/3$) changes the number of adatoms identified quite drastically for the (100) surface, creating many false positives, but also correctly identifies more adatoms on the (110) surface (data not shown).

V. SUMMARY AND CONCLUSIONS

In summary, we have reported recent benchmarks of the continuum-scale drift-diffusion-reaction cluster-dynamics software Xolotl against large-scale atomistic simulations. This study has been conducted within a systematic, hierarchical, multiscale modeling framework to address helium segregation on surfaces of plasma-exposed tungsten, a problem with important implications for the design of PFCs of nuclear fusion reactors. We used a reliable, carefully parameterized interatomic interaction potential and employed a hierarchy of atomic-scale simulations, including MS simulations, targeted MD simulations of near-surface cluster reactions, and large-scale MD simulations of implanted helium evolution in plasma-exposed tungsten to understand the thermodynamic origin of helium surface segregation. This study has also focused on evaluating the effects on helium cluster dynamics of tungsten surface orientation and temperature; performing simulations at two different temperatures, namely, 933 and 1200 K;

and employing reaction rate databases obtained at the foregoing two temperatures.

Our prior atomistic simulations have identified that mobile He_n ($1 \leq n \leq 7$) clusters in the near-surface region are attracted to the surface due to an elastic interaction force that provides a thermodynamic driving force for surface segregation and promotes helium segregation on the surface.^{19–21} Clusters near the surface undergo reactions, most importantly TM, at rates much higher than those of the corresponding reactions in the bulk material, which we have now fully tabulated at 933 and 1200 K. These near-surface cluster dynamics have significant effects on the surface morphology, near-surface defect structures, and amount of helium retained in the material upon plasma exposure; these morphological, structural, and compositional effects are observed in large-scale MD simulations. We integrated the findings of our atomic-scale simulations into a properly parameterized continuum-scale cluster transport–reaction dynamics model, which was tested by comparing its predictions to those of large-scale MD simulations. Therefore, simulations based on our coarse-grained cluster-dynamics model are an efficient computational means for predicting implanted helium evolution, surface segregation, and its near-surface effects in tungsten up to certain levels of helium fluence while additional helium bubble and other high-helium-content physics are being implemented in the coarse-grained model to increase its predictive capabilities at higher fluences required for a full benchmarking comparison to experimental data.

The most important comparisons between the large-scale MD simulation results and the predictions of the Xolotl simulations were carried out under consistent helium implantation conditions. Given that our computations involved single-crystalline tungsten with well-specified surface orientations, i.e., simpler structures and much-better-characterized surfaces than those used in available experimental studies, a systematic comparison of our findings with those reported in experimental works in the literature is well beyond the scope of this paper. However, our findings are generally in qualitative agreement with those of experimental studies with well-characterized samples over a range of implantation conditions. For example, a recent experimental study³⁸ reported helium concentrations under various tungsten surfaces, including “fuzzy” surfaces, as a function of surface temperature and helium fluence. Over the temperature range of 470 to 2595 K and the helium fluence range of 3×10^{24} to $1.3 \times 10^{27} \text{ m}^{-2}$ examined, the measured helium concentration was found to be in the range of 1 to 4 at. %. The helium retention does not exhibit any strong temperature dependence, which is

consistent with our computational findings: The dominant, low-barrier TM reactions retain helium in the near-surface region of tungsten, trapped in the form of He_mV_k complexes, as the small mobile helium clusters undergo drift transport toward the surface.

Our cluster-dynamics model, implemented computationally through Xolotl, is a fully atomistically informed, coarse-grained model that sets the stage for computationally efficient simulations of helium surface segregation, helium retention, and surface morphological evolution toward optimal design of PFCs. The physics of surface segregation, cluster transport, and cluster reactions in the bulk and near the surface have been incorporated into our present cluster-dynamics modeling framework. Extending this framework to include helium bubble physics and all the relevant physical phenomena away from the dilute limit of helium concentration and the impact of such phenomena on surface morphology and near-surface structure is currently underway and will be reported in forthcoming publications, including comparisons to experimental data at a lower helium implantation flux.

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